

IoT BASED SMARTCAMERA

Using Android and Raspberry Pi

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Presented here is a project that lets your Raspberry Pi (RPI) turn into a remote camera and then control and watch live video being captured by this camera on your smartphone from anywhere on

the planet. You just need an Internet connection on your RPI and smartphone. This project explains step-by-step procedure to build your own remote smart security camera.

Hardware required for building the camera are Android smartphone (4.0 or higher), RPI 2, Raspberry camera module, 5V, 1A power adaptor with micro-USB output, USB audio adaptor (optional) and USB Wi-Fi dongle (RPI 2-compatible).

Software required for this project is Butterfly TV app. Since the project is developed using Butterfly TV and RPI, we have named it Butterfly PI. It has the following features:

1. You can broadcast live video in several resolutions.
2. It communicates with your audience via comments while you are broadcasting.
3. It allows you to share live video with your friends and followers on Facebook and Twitter.
4. You can sign up through Facebook or Google +.
5. You can watch and share your videos later and delete whenever you want.

Smartphone setup

Download and install Butterfly TV from Google Play store. Butterfly TV is integrated with the smartphone camera. Login to Butterfly TV and click on My Butterfly PIs item in the menu (Fig. 1). You should see the screen shown in Fig. 2 when you first open My Butterfly PIs screen.

Click on Add new Butterfly PI in the menu as shown in Fig. 3. Write your chosen camera name as device name and security code in Add new Butterfly PI dialogue box. You will

need the name and code later to authenticate RPI. For example, let us select Balboa for both device name and security code (Fig. 4). After clicking OK, you should see your Butterfly PI as offline (Fig. 5).

Raspberry Pi setup

Download the image of Butterfly PI for RPI from <https://drive.google.com/file/d/0BYSlfzRBBmAZkRzSmFEajByWU0/view?ts=5713a68b>

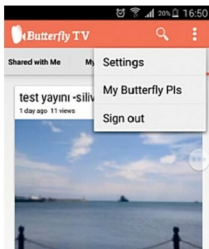


Fig. 1: My Butterfly PIs

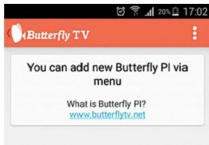


Fig. 2: My Butterfly PIs screen window

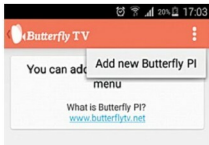


Fig. 3: Add new item

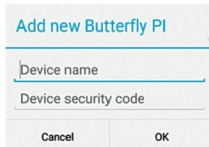


Fig. 4: Device name and security code

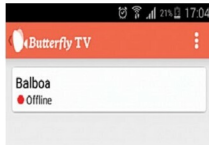


Fig. 5: Butterfly TV offline mode

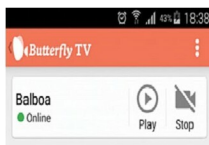


Fig. 6: Butterfly online mode